Neuro Vision OV5640 (-W)

Datasheet



5 MP auto-focus camera expansion board for the Neuro N6, with a VL53L1CB Time-of-Flight ranging sensor and a white illumination LED. The OV5640-W variant adds a u-blox MAYA-W476 module for 2.4 GHz Wi-Fi and BLE.

5 MP
OV5640 image sensor

2 lanes
MIPI CSI-2

72°
Auto-focus lens

Wi-Fi/BLE
OV5640-W variant

PART NUMBER	HARDWARE REVISION	DOCUMENT	DATE
OL-DEV-N6-002 / -003	Rev B	OL-DEV-N6-002 Rev D	25 October 2026

Contents

1 Overview 3
2 Introduction 4
2.1 Variants and ordering information 4
2.2 Related products 4
3 Features 5
4 Mechanical 6
5 Block diagram 8
6 Hardware interfaces 9
6.1 Camera module 9
6.2 Time-of-Flight sensor 9
6.3 Illumination LED 9
6.4 Wireless module (OV5640-W) 9
6.5 Connectors 10
7 Board operation 11
7.1 Absolute maximum ratings 11
7.2 Recommended operating conditions 11
7.3 Electrical characteristics 12
7.4 Power supply 12
7.5 Reset and clocking 13
8 Peripheral interfaces 14
8.1 MIPI CSI-2 14
8.2 I2C 14
8.3 Camera control signals 15
8.4 Time-of-Flight signals 15
8.5 SDIO (OV5640-W) 15
8.6 BLE HCI UART (OV5640-W) 15
8.7 Wireless control signals (OV5640-W) 16
8.8 40-pin board-to-board connector 16
8.9 30-pin board-to-board connector 18
9 Laser safety 19
9.1 Laser classification 19
9.2 Handling precautions 19
10 Regulatory compliance 20
10.1 Product marking 20
10.2 Electromagnetic compatibility 20
10.3 Radio (OV5640-W) 20
10.4 Materials and disposal 21
11 Schematics 22
12 Document revision history 26

1 Overview

The Neuro Vision OV5640 is a 5 megapixel camera expansion board for the Neuro N6. It combines an OmniVision OV5640 sensor with an auto-focus lens, a Time-of-Flight ranging sensor and an illumination LED on a Feather-size board that mounts directly on the Neuro N6.

- OmniVision OV5640 5 MP CMOS image sensor, 2592 x 1944 (QSXGA), 1/4 in optical format, 1.4 µm pixels.
- Auto-focus lens module (72° field of view, 4-element lens with IR filter) with single and continuous auto-focus driven by the OV5640 integrated VCM controller.
- 2-lane MIPI CSI-2 video interface: QSXGA at 15 fps, 1080p at 30 fps, 720p and VGA at 60 fps.
- RGB565 output from the sensor, with JPEG encoding on the Neuro N6 for streaming.
- VL53L1CB Time-of-Flight ranging sensor in long distance mode, ranging a 4 x 4 grid of zones.
- White illumination LED switched on and off from the camera STROBE output.
- Powered from the Neuro N6 3.3 V and 1.8 V rails through three on-board LDO regulators.
- One 30-pin and one 40-pin board-to-board connector, 3.5 mm mating height.
- OV5640-W variant: u-blox MAYA-W476 module with 2.4 GHz Wi-Fi (802.11b/g/n) and BLE.
- Supported by the Neuro N6 Arduino core.

2 Introduction

The Neuro Vision OV5640 mounts on the Neuro N6 through two board-to-board connectors with a mating height of 3.5 mm. Together the two boards form a platform for edge AI vision applications such as object detection, image classification and depth-assisted scene understanding.

The board is available in two variants. The standard Neuro Vision OV5640 provides the camera, depth sensing and illumination functions. The Neuro Vision OV5640-W adds a u-blox MAYA-W476 wireless module with 2.4 GHz Wi-Fi (802.11b/g/n) and BLE, for example to stream video frames over Wi-Fi. The OV5640-W is offered only in countries where its assessed radio configuration is permitted.

The Neuro N6 Arduino core, available from neuron6.com, provides drivers and examples for the camera, the Time-of-Flight sensor and the illumination LED. Unless stated otherwise, the information in this document applies to both variants.

For detailed information on the OV5640 image sensor, the VL53L1CB sensor and the MAYA-W476 module, refer to the OmniVision, STMicroelectronics and u-blox documentation.

2.1 Variants and ordering information

VARIANT	PART NUMBER	HARDWARE REVISION	WIRELESS	DESCRIPTION
Neuro Vision OV5640	`OL-DEV-N6-002`	Rev B	None	5 MP auto-focus camera, Time-of-Flight sensor and illumination LED
Neuro Vision OV5640-W	`OL-DEV-N6-003`	Rev B	2.4 GHz Wi-Fi, BLE	Adds the u-blox MAYA-W476 module. Available in selected countries only

Table 1. Ordering information. Quote the part number and hardware revision when ordering and when contacting Ohm Lab for support.

2.2 Related products

- Neuro N6 (OL-DEV-N6-001)
- Neuro Vision OV5640-W (OL-DEV-N6-003)
- Neuro Vision ST Cam (OL-DEV-N6-004)
- Neuro Vision Thermal (OL-DEV-N6-005)
- Neuro Vision TFT (OL-DEV-N6-006)
- Neuro Vision TFT-W (OL-DEV-N6-007)
- Neuro N6 Enclosure (OL-DEV-N6-009)
- Neuro N6 Debug Adapter (OL-DEV-N6-010)
- B-CAMS-IMX Adapter (OL-DEV-N6-011)

3 Features

FEATURE	DESCRIPTION
Camera	
OV5640 image sensor	5 MP CMOS, 2592 x 1944 (QSXGA), 1/4 in optical format, 1.4 x 1.4 µm pixels, rolling shutter with automatic exposure control, 2-lane MIPI CSI-2 output, SCCB control interface
Video modes	With the Arduino core: VGA 640 x 480 and WVGA 800 x 480 at 60 fps; XGA 1024 x 768 at 30 fps; 720p at 60 fps; 1080p (centre crop) at 30 fps; QSXGA 2592 x 1944 at 15 fps. Rates are capture maximums; lower rates can be set
Output format	RGB565 from the sensor; JPEG encoded on the Neuro N6 for streaming
Image controls	Automatic or manual exposure, automatic or manual gain (1 to 63.9x), automatic white balance, mirror and flip, through the Neuro N6 Arduino core
Lens module	Auto-focus lens with voice-coil motor, 72° field of view, 4-element plastic lens with IR filter, 1/4 in format, distortion below 1%; module 8.5 x 8.5 x 5.0 mm with a 24-pin FPC; single and continuous auto-focus by the OV5640 embedded controller, firmware downloaded over SCCB
Depth sensing	
VL53L1CB Time-of-Flight sensor	Absolute ranging independent of target colour and reflectance; long distance mode with a 4 x 4 grid of zones, one distance per zone; 940 nm Class 1 VCSEL; 27° field of view; interrupt routed to CONN_M (the Arduino core polls the sensor over I2C)
Illumination	
White LED	2835-package white LED switched by a transistor from the OV5640 STROBE output; on or off under software control
Wireless connectivity (OV5640-W only)	
u-blox MAYA-W476	NXP IW610: 2.4 GHz Wi-Fi 802.11b/g/n 1x1, 20 MHz channels, station or access point, WPA2 and WPA3; BLE peripheral and central; SDIO (Wi-Fi) and 4-wire HCI UART (BLE) host interfaces; embedded PCB antenna
Interfaces	
Board-to-board connectors	40-pin and 30-pin HC-PBB40C receptacles, 0.4 mm pitch, 3.5 mm mating height: 2-lane MIPI CSI-2, I2C (3.3 V host side), Time-of-Flight interrupt, NRST; SDIO, HCI UART and wake signals on the OV5640-W
Camera connector	24-pin, 0.5 mm pitch FPC connector
Power	
Supply	3.3 V and 1.8 V from the Neuro N6; three on-board LDO regulators (2.8 V analog, 1.5 V core, 2.8 V Time-of-Flight); ferrite-filtered analog supply with a separate analog ground

Table 2. Neuro Vision OV5640 features.

4 Mechanical

The board follows the Adafruit Feather outline with the four mounting holes on the 45.72 x 17.78 mm pattern. All dimensions are in millimetres and are taken from the PCB design data and the 3D model. The board mounts back to back with the Neuro N6, so the camera side faces away from the Neuro N6.

PCB outline	50.80 x 22.86 mm, corner radius 2.54 mm
PCB	1.6 mm
Height above the PCB	5.2 mm (OV5640 camera module)
Height below the PCB	3.4 mm (board-to-board receptacles)
Overall height	10.2 mm
Stack height on the Neuro N6	17.4 mm from the Neuro N6 battery connector to the top of the expansion board, with a 3.5 mm mating gap
Mounting holes	4 x Ø2.54 mm plated, 45.72 x 17.78 mm pattern, 2.54 mm from the board edges, connected to GND
Board-to-board receptacles	40-pin centred 17.61 mm and 30-pin centred 12.37 mm from the Feather body edge of the Neuro N6 when mated; both on the board centre line

Table 3. Mechanical summary.

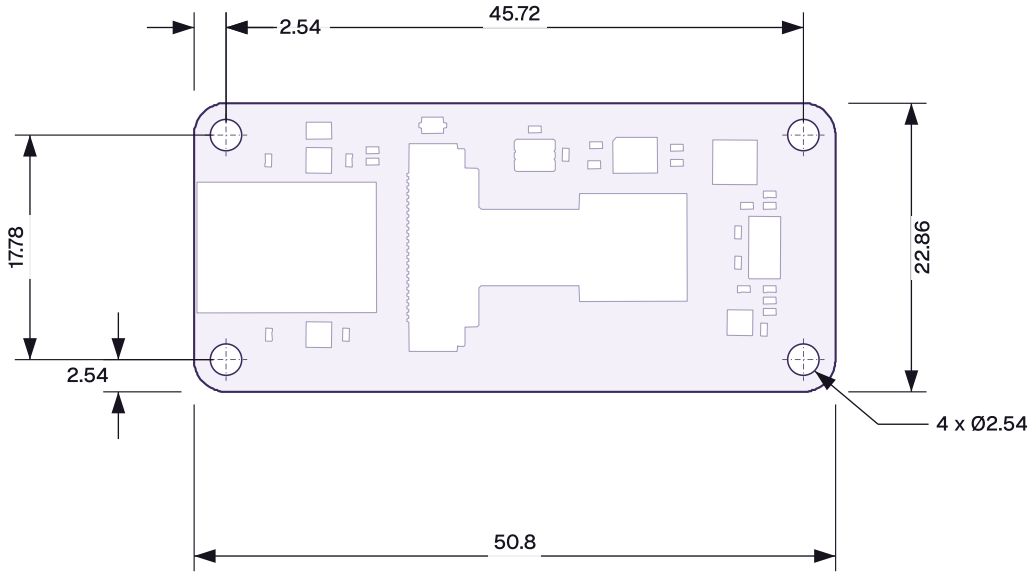


Figure 1. Neuro Vision OV5640 top view (component side). Component outlines are taken from the 3D model; dimensions are nominal design values in millimetres. The wireless module (left) is fitted on the OV5640-W only.

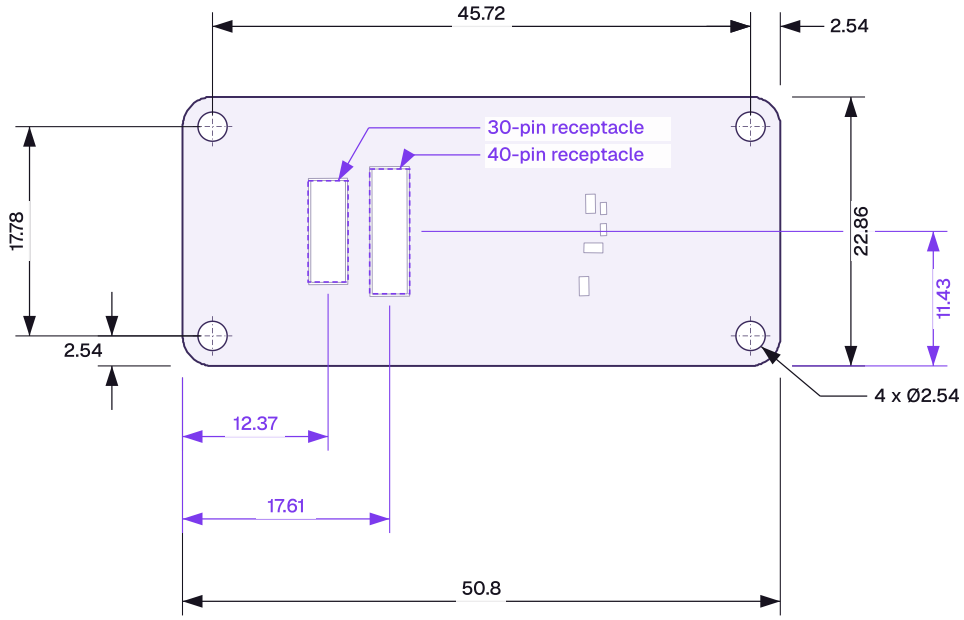


Figure 2. Neuro Vision OV5640 bottom view (mirrored, as seen from below), showing the board-to-board receptacles. Positions match the plug positions on the Neuro N6.

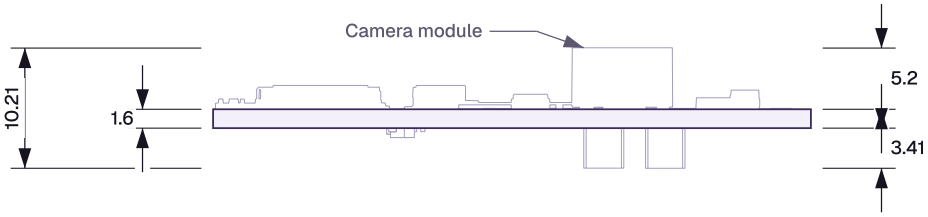


Figure 3. Neuro Vision OV5640 side view.

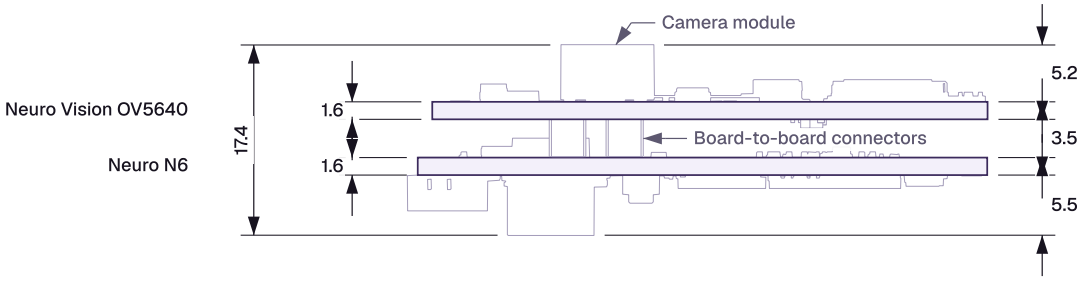


Figure 4. Neuro Vision OV5640 mounted on the Neuro N6, side view to scale from the two 3D models. The boards mount back to back with a 3.5 mm mating gap; the Neuro N6 component side faces down.

5 Block diagram

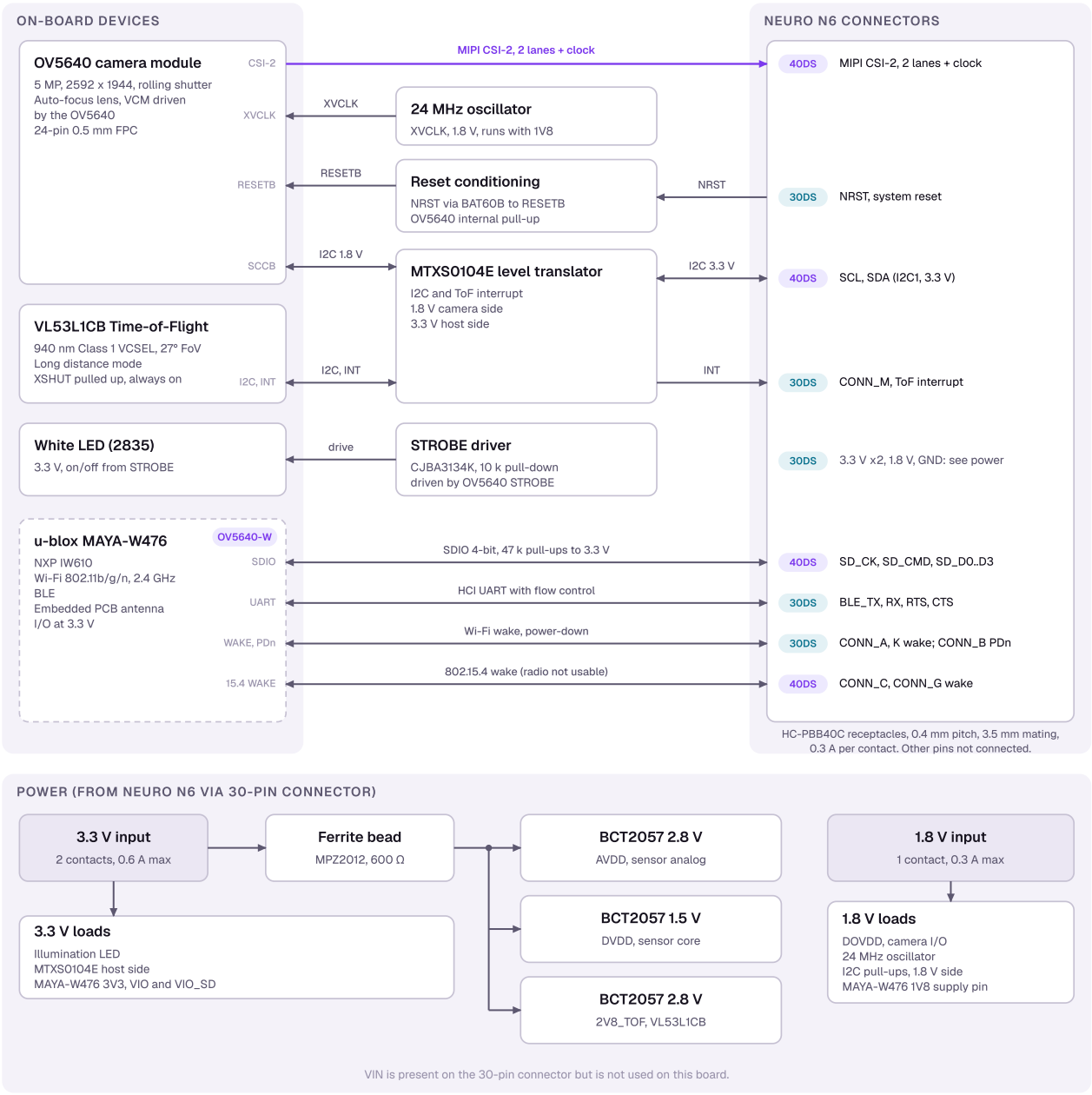


Figure 5. Neuro Vision OV5640 block diagram. The wireless module (dashed) is fitted on the OV5640-W only.

6 Hardware interfaces

6.1 Camera module

The OV5640 camera module connects through a 24-pin, 0.5 mm pitch FPC connector. The module contains the OV5640 sensor and a 72° auto-focus lens assembly driven by the sensor's integrated voice-coil motor controller; the auto-focus firmware is downloaded over the SCCB interface at start-up. An on-board 24 MHz oscillator drives XVCLK, and the sensor is held in reset by the system NRST signal (see the reset and clocking section). Image data leaves the board on the 2-lane MIPI CSI-2 interface routed to the 40-pin connector.

> **Caution** The camera module, its FPC and the FPC connector are sensitive to electrostatic discharge and mechanical stress. Handle the board by its edges and do not connect or disconnect the camera FPC while the board is powered.

6.2 Time-of-Flight sensor

A VL53L1CB ranging sensor is fitted next to the camera module. It emits invisible 940 nm light from an integrated Class 1 VCSEL and measures the photon return time on a SPAD array, giving absolute distance independent of target colour and reflectance over a 27° field of view. The Arduino core runs the sensor in long distance mode and ranges a 4 x 4 grid of zones, one distance per zone. Typical uses alongside the camera are presence detection and attaching absolute distance to detected objects. See the laser safety section for the classification of the emitter and the handling precautions.

The sensor shares the 1.8 V I2C bus with the camera. Its interrupt output (GPIO1) is level-translated to 3.3 V and routed to CONN_M (30-pin connector pin 7) on both variants; the Arduino core polls the sensor over I2C and does not use the interrupt. XSHUT is pulled up to 1.8 V on the board, so the sensor is enabled whenever the board is powered.

6.3 Illumination LED

A high-brightness white LED provides scene illumination. It is powered from the 3.3 V rail and switched by a CJBA3134K transistor driven from the OV5640 STROBE output; a 10 kΩ pull-down holds the LED off while the sensor is in reset. The Arduino core switches the LED on or off through the sensor's SCCB registers. No series resistor is fitted: the LED current is set by the LED forward voltage, the 3.3 V rail and the transistor on-resistance. Take this into account before leaving the LED on for long periods.

6.4 Wireless module (OV5640-W)

The OV5640-W adds a u-blox MAYA-W476 module, a host-based multiradio module built on the NXP IW610 with an embedded PCB antenna. It provides single-stream 2.4 GHz Wi-Fi (802.11b/g/n, 20 MHz channels) and BLE. With the Arduino core the module runs as a Wi-Fi station or access point with WPA2 or WPA3 security, and as a BLE peripheral or central. Wi-Fi traffic uses the Neuro N6 4-bit SDMMC1 interface (SDIO at 3.3 V, up to 50 MHz) and BLE uses the 4-wire USART6 HCI link, so Wi-Fi and BLE use separate host interfaces; the two radios share the 2.4 GHz antenna by time-sharing. The module takes its 3V3, VIO and VIO_SD supplies from the 3.3 V rail, so its I/O is referenced to 3.3 V, and its 1V8 supply from the 1.8 V rail.

Wake-up and control signals are routed to Neuro N6 GPIO: CONN_A and CONN_K for the Wi-Fi wake handshake, CONN_C and CONN_G for the 802.15.4 subsystem wake handshake, and CONN_B for the module power-down input (PDn). The module's 802.15.4 radio is present in silicon, but its SPI host interface is not connected, so it cannot be operated. The configuration pin CON[2] is pulled low with 51 kΩ; CON[0] and CON[1] are left open.

6.5 Connectors

CONNECTOR	PART NUMBER	MATING PART
Camera FPC connector	`FPC-05F-24PH20`	24-way, 0.5 mm pitch FPC (module tail)
30-pin board-to-board receptacle	`HC-PBB40C-30DS-0.4V-3.5-02`	`HC-PBB40C-30DP-0.4V-02` (Neuro N6)
40-pin board-to-board receptacle	`HC-PBB40C-40DS-0.4V-3.5-02`	`HC-PBB40C-40DP-0.4V-02` (Neuro N6)

Table 4. Connector part numbers.

The board-to-board receptacles mate with the plugs fitted to the underside of the Neuro N6 at a mating height of 3.5 mm. The two boards mount back to back, so the component sides of both boards face outwards. Align both connectors before pressing the boards together, and separate the boards evenly when removing the expansion board.

7 Board operation

7.1 Absolute maximum ratings

Absolute maximum ratings are stress ratings only. They are not operating conditions, and functional operation at or beyond these limits is neither implied nor guaranteed. Exposure to conditions at the limits may reduce reliability. Each limit is the lowest rating among the components connected to that net on the Neuro Vision OV5640 schematic. All voltages are referenced to GND.

PARAMETER	MIN	MAX	UNIT	LIMITING ELEMENT
Supply voltage, VDD_3V3 (OV5640)	-0.3	6.0	V	MTXS0104E supply (VCCB)
Supply voltage, VDD_3V3 (OV5640-W)	-0.3	3.96	V	MAYA-W476 3V3 supply input
Supply voltage, VDD_1V8 (OV5640)	-0.3	3.6	V	VL53L1CB I/O (pull-ups to 1V8)
Supply voltage, VDD_1V8 (OV5640-W)	-0.3	2.16	V	MAYA-W476 1V8 supply input
Voltage on host-side 3.3 V signals (SCL, SDA, CONN_M)	-0.3	VDD_3V3 + 0.3	V	MTXS0104E B port
Voltage on NRST	-0.3	VDD_3V3 + 0.3	V	Neuro N6 reset line at 3.3 V; BAT60B on this board
Voltage on camera-domain 1.8 V signals (SCL1V8, SDA1V8, XCLK)	-0.3	VDD_1V8 + 0.3	V	MTXS0104E A port, VL53L1CB, oscillator
Voltage on SDIO, HCI UART and control pins (OV5640-W)	-0.3	VIO + 0.4	V	MAYA-W476 input high level maximum (VIO = 3.3 V)
Continuous current per board-to-board contact		0.3	A	HC-PBB40C 0.4 mm pitch contact
Illumination LED forward current		60	mA	XL-2835UWC-02
Storage temperature	-40	85	°C	Component ratings

Table 5. Absolute maximum ratings.

The 3.3 V rail reaches the board on two contacts of the 30-pin connector and the 1.8 V rail on one, giving 0.6 A and 0.3 A respectively at the 0.3 A per-contact limit. The VIN pins on the 30-pin connector are not connected on this board.

7.2 Recommended operating conditions

PARAMETER	MIN	MAX	UNIT	CONDITIONS
Ambient temperature	0	30	°C	Indoor, dry locations only
Relative humidity	25	75	%	Non-condensing
Supply				From the Neuro N6, powered with 5 V DC via USB-C or its battery

Table 6. Recommended operating conditions.

7.3 Electrical characteristics

PARAMETER	MIN	TYP	MAX	UNIT	CONDITIONS
Supply voltage, VDD_3V3	3.23	3.33	3.44	V	From the Neuro N6
Supply voltage, VDD_1V8	1.75	1.8	1.85	V	From the Neuro N6
On-board analog rail, AVDD		2.8		V	BCT2057, sensor analog supply
On-board core rail, DVDD		1.5		V	BCT2057, sensor core supply
On-board Time-of-Flight rail, 2V8_TOF		2.8		V	BCT2057
Camera-domain logic level		1.8		V	DOVDD, I2C, XVCLK
Host-side logic level		3.3		V	I2C, interrupt, SDIO, UART, control

Table 7. Electrical characteristics.

A Neuro N6 with an expansion board typically draws 150 to 500 mA from its 5 V USB supply, depending on the workload; see the Neuro N6 datasheet.

7.4 Power supply

The board has no power input of its own. It is powered from the Neuro N6 through the board-to-board connectors, which provide the 3.3 V (VDD_3V3) and 1.8 V (VDD_1V8) rails. The 3.3 V rail is filtered by a ferrite bead and feeds three BCT2057 low-dropout regulators: one generates the 2.8 V analog supply (AVDD) for the image sensor with a separate analog ground, one generates the 1.5 V digital core supply (DVDD), and one generates an independent 2.8 V supply for the Time-of-Flight sensor and its VCSEL. The camera I/O domain (DOVDD), the 24 MHz oscillator and the I2C pull-ups run directly from the 1.8 V rail. On the OV5640-W the wireless module takes its 3V3, VIO and VIO_SD supplies from the 3.3 V rail, so its I/O is referenced to 3.3 V, and its 1V8 supply from the 1.8 V rail.

RAIL	SOURCE	VOLTAGE	SUPPLIES
`VDD_3V3`	Neuro N6, 30-pin connector	3.3 V	Illumination LED, I2C level translator, LDO inputs (via ferrite bead), MAYA-W476 3V3, VIO and VIO_SD (OV5640-W)
`VDD_1V8`	Neuro N6, 30-pin connector	1.8 V	Camera I/O (DOVDD), 24 MHz oscillator, I2C pull-ups, MAYA-W476 1V8 (OV5640-W)
`AVDD`	On-board LDO	2.8 V	Camera analog supply, separate analog ground
`DVDD`	On-board LDO	1.5 V	Camera digital core
`2V8_TOF`	On-board LDO	2.8 V	Time-of-Flight sensor and VCSEL
`VIN`	Neuro N6, 30-pin connector	VBUS or VBAT	Not used on this board

Table 8. Power rails.

> **Warning** Do not attach or detach the Neuro Vision OV5640 while the Neuro N6 is powered. Disconnect the USB-C supply and the battery before changing expansion boards; otherwise the boards may be permanently damaged.

7.5 Reset and clocking

The system reset signal NRST from the Neuro N6 resets the camera: asserting NRST pulls the sensor's active-low RESETB input low through a BAT60B diode, and RESETB returns high through the sensor's internal pull-up when NRST is released, so the camera is reset whenever the system is reset. The Time-of-Flight sensor is not connected to NRST; it is reset over I2C or by removing board power. The 24 MHz oscillator runs whenever the 1.8 V rail is present and drives XVCLK directly, so no clock is required from the host.

8 Peripheral interfaces

Connector pins are given as `40DS:n` and `30DS:n`, the receptacle pin numbers on this board. The matching Neuro N6 plug pin is listed in the connector tables. The Neuro N6 datasheet gives the MCU pin for every signal.

8.1 MIPI CSI-2

The OV5640 2-lane MIPI CSI-2 interface is routed differentially from the camera FPC connector to the 40-pin connector, matching the Neuro N6 D-PHY receiver (CSI-2 version 1.3 compatible). Ground guard pins separate the lanes.

NET	CONNECTOR PIN	DESCRIPTION
`CSI_D0P`	`40DS:31`	Data lane 0 positive
`CSI_D0N`	`40DS:29`	Data lane 0 negative
`CSI_CLKP`	`40DS:23`	Clock lane positive
`CSI_CLKN`	`40DS:21`	Clock lane negative
`CSI_D1P`	`40DS:15`	Data lane 1 positive
`CSI_D1N`	`40DS:13`	Data lane 1 negative

Table 9. MIPI CSI-2 pins.

8.2 I2C

A single I2C bus controls the camera and the Time-of-Flight sensor. It enters the board at 3.3 V on the 40-pin connector, from the Neuro N6 I2C1 bus that also serves the on-board IMU and magnetometer of the Neuro N6, and is level-translated by an MTXS0104E to the 1.8 V camera domain, where it is pulled up through 2.2 kΩ resistors. Standard mode (100 kHz) and fast mode (400 kHz) are supported.

NET	LOGIC LEVEL	CONNECTOR PIN	DESCRIPTION
`SCL`	3.3 V	`40DS:35`	I2C clock, host side
`SDA`	3.3 V	`40DS:37`	I2C data, host side
`SCL1V8`	1.8 V		I2C clock, camera domain
`SDA1V8`	1.8 V		I2C data, camera domain

Table 10. I2C pins.

DEVICE	7-BIT ADDRESS	NOTES
OV5640 image sensor (SCCB)	`0x3C`	Write 0x78, read 0x79
VL53L1CB Time-of-Flight sensor	`0x29`	Default address, can be reprogrammed at run time

Table 11. I2C addresses.

8.3 Camera control signals

NET	OV5640 PIN	CONNECTOR PIN	DESCRIPTION
XCLK	XVCLK		24 MHz master clock from the on-board oscillator
NRST	RESETB	30DS:29	Active-low camera reset, driven by the system NRST
STROBE	STROBE		Strobe output, switches the white LED

Table 12. Camera control signals.

8.4 Time-of-Flight signals

NET	VL53L1CB PIN	CONNECTOR PIN	DESCRIPTION
CONN_M	GPIO1	30DS:7	Ranging interrupt at 3.3 V (both variants); not used by the Arduino core, which polls the sensor
	XSHUT		Pulled up to 1.8 V on the board; the sensor is always enabled

Table 13. Time-of-Flight signals.

8.5 SDIO (OV5640-W)

On the OV5640-W the Neuro N6 4-bit SDMMC1 interface connects to the module SDIO interface at 3.3 V, up to 50 MHz, and carries all Wi-Fi traffic. The command and data lines are pulled up to 3.3 V through 47 kΩ resistors. On the standard variant these pins are not connected.

NET	MAYA-W476 PAD	CONNECTOR PIN	DESCRIPTION
SD_CK	SD_CLK (A4)	40DS:5	SDIO clock
SD_CMD	SD_CMD (B4)	40DS:3	SDIO command
SD_D0	SD_DAT[0] (B3)	40DS:8	SDIO data 0
SD_D1	SD_DAT[1] (A3)	40DS:1	SDIO data 1
SD_D2	SD_DAT[2] (B5)	40DS:39	SDIO data 2
SD_D3	SD_DAT[3] (A5)	40DS:7	SDIO data 3

Table 14. SDIO pins (OV5640-W).

8.6 BLE HCI UART (OV5640-W)

BLE traffic uses the Neuro N6 4-wire USART6 interface, connected to the module HCI UART. Signal names are given from the host's point of view.

NET	MAYA-W476 PAD	CONNECTOR PIN	DESCRIPTION
BLE_TX	UART_SIN (G8)	30DS:3	Host transmit to module receive
BLE_RX	UART_SOUT (G9)	30DS:5	Module transmit to host receive
BLE_RTS	UART_CTS (H8)	30DS:9	Host request to send
BLE_CTS	UART_RTS (H9)	30DS:8	Host clear to send

Table 15. HCI UART pins (OV5640-W).

8.7 Wireless control signals (OV5640-W)

NET	MAYA-W476 PAD	CONNECTOR PIN	DESCRIPTION
CONN_A	WL_WAKE_OUT / JTAG_TDI (F7)	30DS:30	Wi-Fi wake-up, module to host
CONN_K	WL_WAKE_IN (J8)	30DS:20	Wi-Fi wake-up, host to module
CONN_C	NB_WAKE_OUT / JTAG_TDO (E7)	40DS:38	802.15.4 subsystem wake-up, module to host
CONN_G	NB_WAKE_IN (J9)	40DS:12	802.15.4 subsystem wake-up, host to module
CONN_B	PDn (J3)	30DS:22	Module power-down input, active low

Table 16. Wireless control signals (OV5640-W). CON[2] is pulled low on the board; CON[0] and CON[1] are open.

8.8 40-pin board-to-board connector

PIN	NET	N6 PLUG PIN	FUNCTION
1	SD_D1	40	SDIO data 1 (OV5640-W only)
2	LCD_R3	39	LCD interface (not used on this board)
3	SD_CMD	38	SDIO command (OV5640-W only)
4	LCD_R7	37	LCD interface (not used on this board)
5	SD_CK	36	SDIO clock (OV5640-W only)
6	LCD_R6	35	LCD interface (not used on this board)
7	SD_D3	34	SDIO data 3 (OV5640-W only)
8	SD_D0	33	SDIO data 0 (OV5640-W only)
9	GND	32	Ground
10	LCD_VSYNC	31	LCD interface (not used on this board)
11	GND	30	Ground
12	CONN_G	29	802.15.4 wake-up input (OV5640-W only)
13	CSI_D1N	28	MIPI CSI-2 data lane 1 negative
14	LCD_CLK	27	LCD interface (not used on this board)
15	CSI_D1P	26	MIPI CSI-2 data lane 1 positive
16	LCD_B5	25	LCD interface (not used on this board)
17	GND	24	Ground
18	LCD_B3	23	LCD interface (not used on this board)
19	GND	22	Ground
20	LCD_B4	21	LCD interface (not used on this board)
21	CSI_CLKN	20	MIPI CSI-2 clock lane negative
22	LCD_B6	19	LCD interface (not used on this board)
23	CSI_CLKP	18	MIPI CSI-2 clock lane positive
24	LCD_B7	17	LCD interface (not used on this board)

PIN	NET	N6 PLUG PIN	FUNCTION
25	GND	16	Ground
26	CONN_F	15	GPIO (not used on this board)
27	GND	14	Ground
28	LCD_G5	13	LCD interface (not used on this board)
29	CSI_D0N	12	MIPI CSI-2 data lane 0 negative
30	LCD_G3	11	LCD interface (not used on this board)
31	CSI_D0P	10	MIPI CSI-2 data lane 0 positive
32	LCD_G7	9	LCD interface (not used on this board)
33	GND	8	Ground
34	LCD_G6	7	LCD interface (not used on this board)
35	SCL	6	I2C clock, 3.3 V host side (Neuro N6 I2C1)
36	LCD_R4	5	LCD interface (not used on this board)
37	SDA	4	I2C data, 3.3 V host side (Neuro N6 I2C1)
38	CONN_C	3	802.15.4 wake-up output (OV5640-W only)
39	SD_D2	2	SDIO data 2 (OV5640-W only)
40	CONN_D	1	GPIO (not used on this board)

Table 17. 40-pin receptacle pin assignment. On the standard variant the SDIO and wake pins are not connected.

8.9 30-pin board-to-board connector

PIN	OV5640	OV5640-W	N6 PLUG PIN	FUNCTION
1	LCD_HSYNC	LCD_HSYNC	29	LCD interface (not used on this board)
2	SAI1_MCLK_B	SAI1_MCLK_B	30	Audio interface (not used on this board)
3	CONN_J	BLE_TX	27	OV5640-W: BLE UART host transmit
4	LCD_R5	LCD_R5	28	LCD interface (not used on this board)
5	CONN_H	BLE_RX	25	OV5640-W: BLE UART host receive
6	CONN_N	Not connected	26	Not used on this board
7	CONN_M	CONN_M	23	Time-of-Flight interrupt to the host (both variants)
8	CONN_L	BLE_CTS	24	OV5640-W: BLE UART host clear to send
9	CONN_E	BLE_RTS	21	OV5640-W: BLE UART host request to send
10	DEBUG_CLK	DEBUG_CLK	22	SWD clock (not used on this board)
11	LCD_DE	LCD_DE	19	LCD interface (not used on this board)
12	DEBUG_DAT	DEBUG_DAT	20	SWD data (not used on this board)
13	VIN	VIN	17	VIN rail from the Neuro N6 (not used on this board)
14	LCD_G4	LCD_G4	18	LCD interface (not used on this board)
15	VIN	VIN	15	VIN rail from the Neuro N6 (not used on this board)
16	SAI1_SCK_B	SAI1_SCK_B	16	Audio interface (not used on this board)
17	VIN	VIN	13	VIN rail from the Neuro N6 (not used on this board)
18	SAI1_SD_B	SAI1_SD_B	14	Audio interface (not used on this board)
19	VIN	VIN	11	VIN rail from the Neuro N6 (not used on this board)
20	CONN_K	CONN_K	12	OV5640-W: Wi-Fi wake-up, host to module
21	VIN	VIN	9	VIN rail from the Neuro N6 (not used on this board)
22	CONN_B	CONN_B	10	OV5640-W: module power-down (PDn)
23	3V3	3V3	7	3.3 V supply input
24	GND	GND	8	Ground
25	3V3	3V3	5	3.3 V supply input
26	LCD_G2	LCD_G2	6	LCD interface (not used on this board)
27	1V8	1V8	3	1.8 V supply input
28	SAI1_FS_B	SAI1_FS_B	4	Audio interface (not used on this board)
29	NRST	NRST	1	System reset input (resets the camera)
30	CONN_A	CONN_A	2	OV5640-W: Wi-Fi wake-up, module to host

Table 18. 30-pin receptacle pin assignment.

9 Laser safety

The Neuro Vision OV5640 and OV5640-W contain a laser emitter. The VL53L1CB Time-of-Flight sensor uses an integrated vertical-cavity surface-emitting laser (VCSEL) and emits invisible 940 nm infrared light through the optical window on the top face of the sensor package while a ranging measurement is in progress. No other component on the board emits laser radiation.

9.1 Laser classification

The VL53L1CB is a Class 1 laser product and is safe under all conditions of reasonably foreseeable normal use. No protective eyewear or other protective measures are required, provided the board is used as described in this document.

Laser source	VCSEL integrated within the VL53L1CB
Laser class	Class 1
Classification standard	IEC 60825-1:2014 (Edition 3.0)
Wavelength	940 nm, invisible infrared
Field of illumination	27° full field of view
Aperture	Optical window on the top face of the VL53L1CB package



Figure 6. Class 1 laser product label, as printed on the leaflet supplied with the board.

9.2 Handling precautions

- The emitted light is invisible. Do not assume the emitter is inactive because no light can be seen: the VCSEL is enabled whenever the host has started a ranging measurement.
- Do not examine the sensor window at close range through magnifying optics such as an eye loupe, a magnifier or a microscope.
- Do not fit additional lenses, mirrors or other optical elements in front of the sensor. External optics can concentrate the beam and invalidate the Class 1 classification.
- Do not operate a board on which the VL53L1CB package or its optical window is cracked, scratched, contaminated or otherwise damaged.
- Do not rework, remove or replace the VL53L1CB. The sensor contains no user-serviceable parts, and disassembly may expose the emitter without its optical window.
- Configure the sensor through the ST VL53L1CB API or the Neuro N6 Arduino core drivers, and keep ranging parameters within the ranges given in the ST documentation.

The white illumination LED is not a laser product and falls outside the scope of IEC 60825-1. It is nevertheless a high-brightness source: avoid looking directly into the LED at close range while it is switched on.

Where the Neuro Vision OV5640 or OV5640-W is built into an end product, the manufacturer of that product is responsible for its compliance with IEC 60825-1, including any labelling and user information.

10 Regulatory compliance

This section describes the regulatory status of the Neuro Vision OV5640 and Neuro Vision OV5640-W as supplied by Ohm Lab. The board is an expansion board intended to be operated only when mounted on a Neuro N6, and it is assessed as part of that combination rather than as a stand-alone product. Where the board is incorporated into a finished product, responsibility for the compliance of that product rests with its manufacturer.

10.1 Product marking

SYMBOL	MARKING	APPLIED TO	MEANING
CE	CE	Board and packaging	Conformity with the EMC Directive 2014/30/EU (OV5640) or the Radio Equipment Directive 2014/53/EU (OV5640-W), and with the RoHS Directive 2011/65/EU.
	Crossed-out wheeled bin	Board and packaging	Do not dispose of with unsorted household waste. The bar identifies equipment placed on the market after 13 August 2005.
	Triman and Info-tri	Packaging	France: sorting information required by the French environment code. The Triman is printed on the carton; an Info-tri sticker with the sorting instructions is added to orders for France.
	Class 1 laser product	Leaflet	Explanatory label required by IEC 60825-1:2014 for the VL53L1CB emitter, printed on the leaflet supplied with the board. See the laser safety section.

Table 19. Product markings. The controlled artwork is held in the technical file.

No UKCA marking is applied: CE marking is accepted for goods placed on the Great Britain market. RoHS conformity is declared through the CE marking and carries no separate mark.

10.2 Electromagnetic compatibility

The Neuro N6 platform, including the Neuro Vision OV5640 and OV5640-W, has been assessed for electromagnetic compatibility as a product family, based on tests of worst-case configurations of the Neuro N6 with expansion boards fitted. The Neuro Vision OV5640 conforms to the EMC Directive 2014/30/EU, with emissions assessed to EN 55032 Class B and immunity to EN 55035. The OV5640-W is radio equipment, so its electromagnetic compatibility is covered by the Radio Equipment Directive 2014/53/EU. The radio information required by that directive is supplied with the board. The assessment applies to the board mounted on a Neuro N6 and operated with the default clock and radio settings of the Neuro N6 Arduino core. The declarations of conformity are published at neuron6.com/conformity.

10.3 Radio (OV5640-W)

The Neuro Vision OV5640-W contains a u-blox MAYA-W476 module providing 2.4 GHz Wi-Fi (802.11b/g/n) and BLE. The standard Neuro Vision OV5640 contains no radio transmitter.

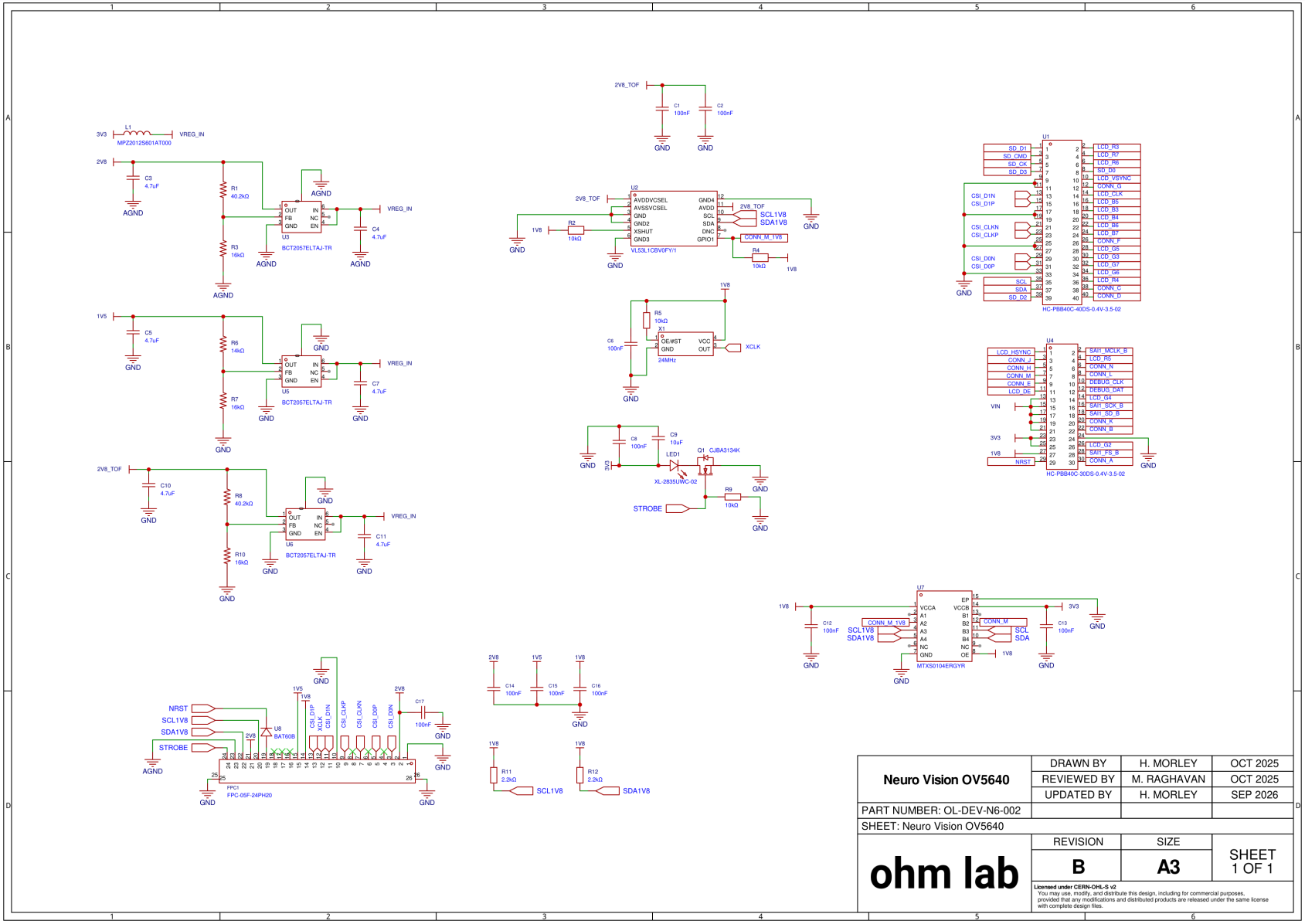
10.4 Materials and disposal

Both variants of the Neuro Vision OV5640 are manufactured to the substance restrictions of Directive 2011/65/EU (RoHS 2) as amended and the UK Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012. Supplier material declarations, and declarations covering substances of very high concern under REACH, are held in the technical file.

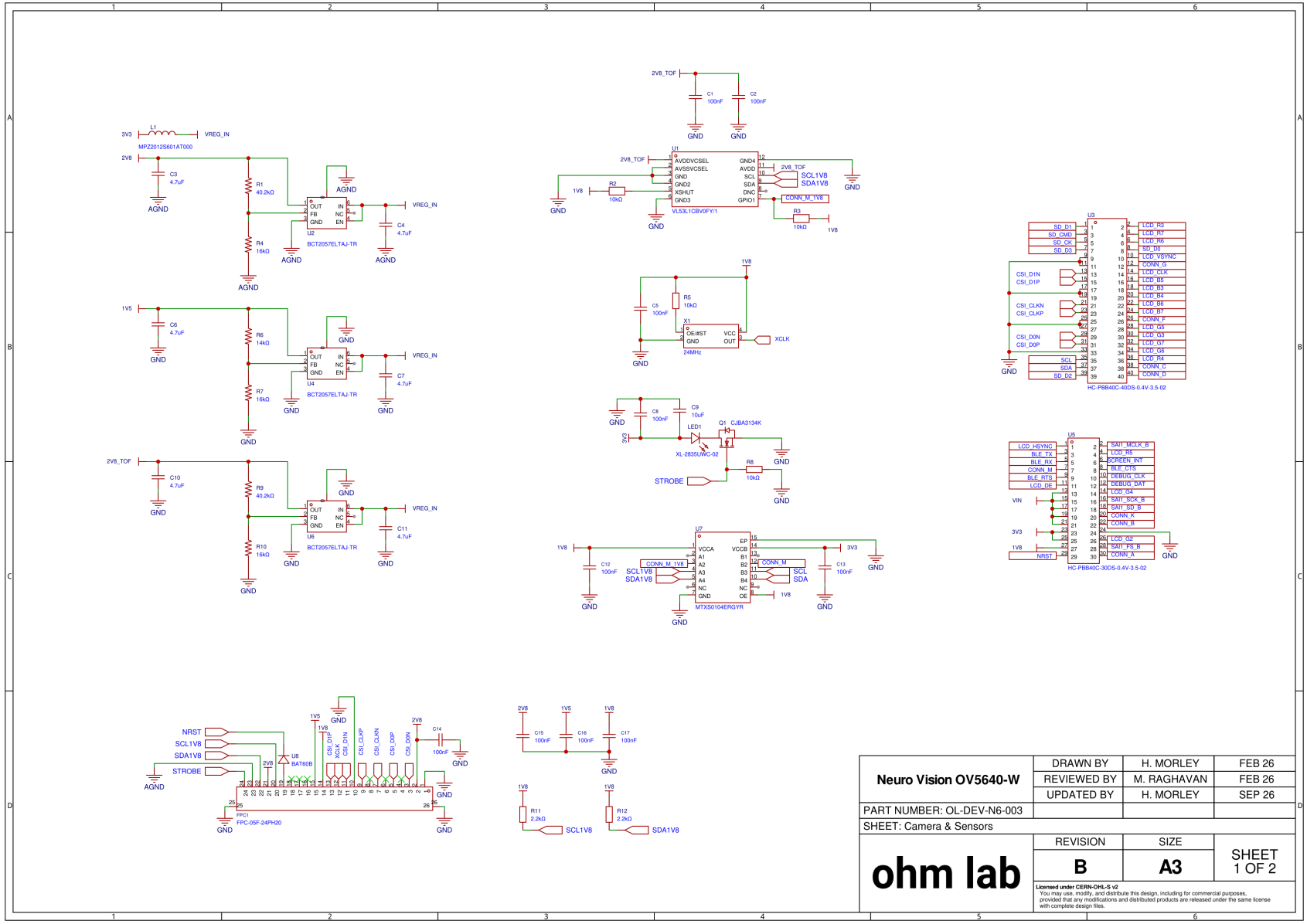
At end of life, do not dispose of the board with unsorted household waste. Return it to a designated collection point for electrical and electronic equipment, or to Ohm Lab.

11 Schematics

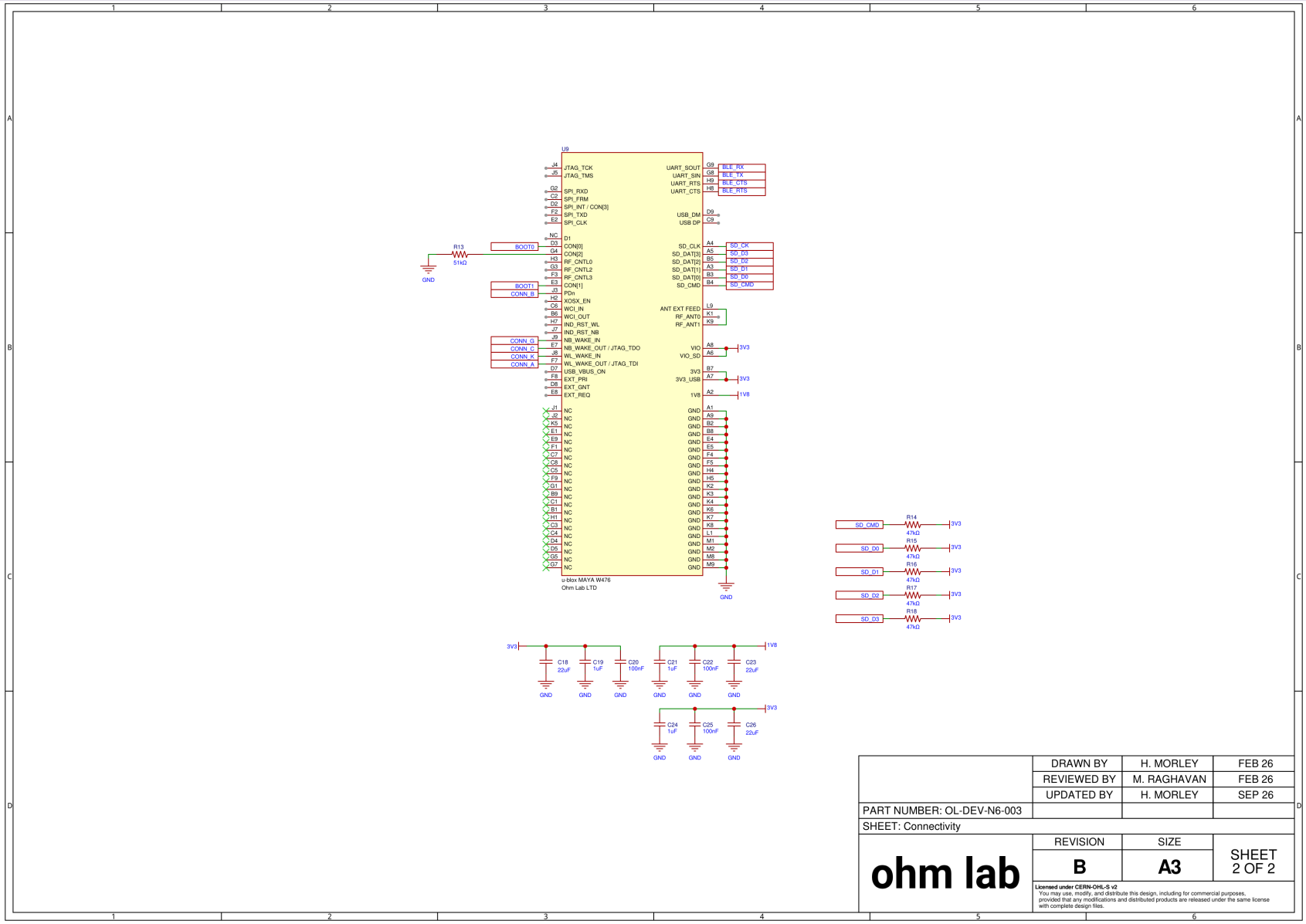
The schematics are reproduced on the following pages for reference. Net names match those used in this document. Full-resolution sheets are published in the Neuro N6 Hardware repository (github.com/ohmlab-ltd/Neuro-N6-Hardware).



Neuro Vision OV5640 Rev B schematic: camera, Time-of-Flight sensor, illumination, power and connectors.



Neuro Vision OV5640-W Rev B schematic, sheet 1 of 2: camera and sensors.



Neuro Vision OV5640-W Rev B schematic, sheet 2 of 2: connectivity (MAYA-W476 module and connectors).

12 Document revision history

REVISION	DATE	CHANGES
A	02/07/2026	• Pre-production first release
B	02/09/2026	• Ratings, laser safety and regulatory section added
C	24/09/2026	• Rev B hardware • New block diagram and drawings
D	25/10/2026	• Title covers both variants • Features as supported by the Arduino core • Wi-Fi 2.4 GHz only

Table 20. Document revision history.